$$\lambda_3 \to \frac{1}{4}\left(-g_1^2+g_2^2\right)+\frac{1}{16\pi^2}$$
$$\Big(\frac{1}{12}(-6g_1^2g_2^2+7g_2^4)+\frac{1}{12}\sum_p g_1^4\,\mathrm{LF}_{2,0}[m_{\tilde d}^{p}]-\frac{1}{2}g_1^2\,\overline{y_d}^{pr}y_d^{pr}\,\mathrm{LF}_{2,0}[m_{\tilde d}^{r}]+\frac{1}{4}\sum_p g_1^4\,\mathrm{LF}_{2,0}[m_{\tilde e}^{p}]-$$
$$\frac{1}{2}g_1^2\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{2,0}[m_{\tilde e}^{r}]+\frac{1}{8}(g_1^2+g_2^2)\Big(2\,\overline{y_e}^{pr}y_e^{pr}+\sum_p(g_1^2-g_2^2)\Big)\mathrm{LF}_{2,0}[m_{\tilde l}^{p}]+$$
$$\frac{1}{24}\Big(-6\,\overline{y_d}^{pr}y_d^{pr}(g_1^2-3g_2^2)+6\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+3g_2^2)+\sum_p(g_1^4-9g_2^4)\Big)\mathrm{LF}_{2,0}[m_{\tilde q}^{p}]+$$
$$\frac{1}{3}\sum_p g_1^4\,\mathrm{LF}_{2,0}[m_{\tilde u}^{p}]-g_1^2\,\overline{y_u}^{pr}y_u^{pr}\,\mathrm{LF}_{2,0}[m_{\tilde u}^{r}]+\frac{1}{2}g_1^4\,\mathrm{LF}_{2,2,-2}[m_1,\tilde\mu]+$$
$$\frac{1}{2}g_1^4m_1^2\,\mathrm{LF}_{2,2,-1}[m_1,\tilde\mu]+\frac{5}{2}g_2^4\,\mathrm{LF}_{2,2,-2}[m_2,\tilde\mu]+\frac{1}{2}g_2^4m_2^2\,\mathrm{LF}_{2,2,-1}[m_2,\tilde\mu]+$$
$$\frac{2}{3}m_2\tilde\mu g_2^4(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})\mathrm{LF}_{2,2,0}[m_2,\tilde\mu]-\frac{1}{3}m_2\tilde\mu g_2^4(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})\mathrm{LF}_{3,2,-1}[m_2,\tilde\mu]+$$
$$\frac{1}{2}g_1^2(-\overline{a_d}^{pr}a_d^{pr}+\tilde\mu^2\,\overline{y_d}^{pr}y_d^{pr})\mathrm{LF}_{2,1,0}[m_{\tilde d}^{r},m_{\tilde q}^{p}]+$$
$$\frac{1}{2}g_1^2(-C_{\phi2^2}\overline{a_d}^{pr}a_d^{pr}+C_{\phi1^2}\tilde\mu^2\,\overline{y_d}^{pr}y_d^{pr})\mathrm{LF}_{2,2,0}[m_{\tilde d}^{r},m_{\tilde q}^{p}]-$$
$$3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,\mathrm{LF}_{1,1,0}[m_{\tilde d}^{r},m_{\tilde u}^{t}]+\frac{1}{2}g_1^2(-\overline{a_e}^{pr}a_e^{pr}+\tilde\mu^2\,\overline{y_e}^{pr}y_e^{pr})\mathrm{LF}_{2,1,0}[m_{\tilde e}^{r},m_{\tilde l}^{p}]+$$
$$\frac{1}{2}g_1^2(-C_{\phi2^2}\overline{a_e}^{pr}a_e^{pr}+C_{\phi1^2}\tilde\mu^2\,\overline{y_e}^{pr}y_e^{pr})\mathrm{LF}_{2,2,0}[m_{\tilde e}^{r},m_{\tilde l}^{p}]+$$
$$\frac{1}{2}(g_2^2\,\overline{a_e}^{pr}a_e^{pr}-g_1^2\tilde\mu^2\,\overline{y_e}^{pr}y_e^{pr})\mathrm{LF}_{2,1,0}[m_{\tilde l}^{p},m_{\tilde e}^{r}]+\frac{1}{4}(g_1^2-g_2^2)(\overline{a_e}^{pr}a_e^{pr}+\tilde\mu^2\,\overline{y_e}^{pr}y_e^{pr})$$
$$\mathrm{LF}_{3,1,-1}[m_{\tilde l}^{p},m_{\tilde e}^{r}]+\frac{1}{2}\Big(\tilde\mu\,\overline{y_e}^{pr}\big(\overline{C_{\phi1\phi2}}\,a_e^{pr}(-g_1^2+g_2^2)-2C_{\phi1^2}\tilde\mu g_1^2y_e^{pr}\big)+$$
$$\overline{a_e}^{pr}\big(2C_{\phi2^2}g_2^2a_e^{pr}+C_{\phi1\phi2}\tilde\mu y_e^{pr}(-g_1^2+g_2^2)\big)\Big)\mathrm{LF}_{3,1,0}[m_{\tilde l}^{p},m_{\tilde e}^{r}]+$$
$$\frac{1}{2}g_1^2(C_{\phi2^2}\overline{a_e}^{pr}a_e^{pr}-C_{\phi1^2}\tilde\mu^2\,\overline{y_e}^{pr}y_e^{pr})\mathrm{LF}_{3,2,-1}[m_{\tilde l}^{p},m_{\tilde e}^{r}]+$$
$$\frac{1}{4}\Big(\overline{a_e}^{pr}\big(3C_{\phi2^2}a_e^{pr}(g_1^2-3g_2^2)+C_{\phi1\phi2}\tilde\mu y_e^{pr}(6g_1^2-5g_2^2)\big)+$$
$$\tilde\mu\,\overline{y_e}^{pr}\big(\overline{C_{\phi1\phi2}}\,a_e^{pr}(6g_1^2-5g_2^2)+3C_{\phi1^2}\tilde\mu y_e^{pr}(3g_1^2-g_2^2)\big)\Big)\mathrm{LF}_{4,1,-1}[m_{\tilde l}^{p},m_{\tilde e}^{r}]+$$
$$\frac{1}{3}\Big(\tilde\mu\,\overline{y_e}^{pr}\big(\overline{C_{\phi1\phi2}}\,a_e^{pr}(-3g_1^2+2g_2^2)+3C_{\phi1^2}\tilde\mu y_e^{pr}(-g_1^2+g_2^2)\big)+$$
$$\overline{a_e}^{pr}\big(3C_{\phi2^2}a_e^{pr}(-g_1^2+g_2^2)+C_{\phi1\phi2}\tilde\mu y_e^{pr}(-3g_1^2+2g_2^2)\big)\Big)\mathrm{LF}_{5,1,-2}[m_{\tilde l}^{p},m_{\tilde e}^{r}]+$$
$$\frac{1}{2}(\overline{a_d}^{pr}a_d^{pr}(-2g_1^2+3g_2^2)-g_1^2\tilde\mu^2\,\overline{y_d}^{pr}y_d^{pr})\mathrm{LF}_{2,1,0}[m_{\tilde q}^{p},m_{\tilde d}^{r}]+$$
$$\frac{3}{4}(g_1^2-g_2^2)(\overline{a_d}^{pr}a_d^{pr}+\tilde\mu^2\,\overline{y_d}^{pr}y_d^{pr})\mathrm{LF}_{3,1,-1}[m_{\tilde q}^{p},m_{\tilde d}^{r}]+$$
$$\frac{1}{2}\Big(\tilde\mu\,\overline{y_d}^{pr}\big(3\,\overline{C_{\phi1\phi2}}\,a_d^{pr}(-g_1^2+g_2^2)-2C_{\phi1^2}\tilde\mu g_1^2y_d^{pr}\big)+$$
$$\overline{a_d}^{pr}\big(2C_{\phi2^2}a_d^{pr}(-2g_1^2+3g_2^2)+3C_{\phi1\phi2}\tilde\mu y_d^{pr}(-g_1^2+g_2^2)\big)\Big)\mathrm{LF}_{3,1,0}[m_{\tilde q}^{p},m_{\tilde d}^{r}]+$$
$$\frac{1}{2}g_1^2(C_{\phi2^2}\overline{a_d}^{pr}a_d^{pr}-C_{\phi1^2}\tilde\mu^2\,\overline{y_d}^{pr}y_d^{pr})\mathrm{LF}_{3,2,-1}[m_{\tilde q}^{p},m_{\tilde d}^{r}]+$$
$$\frac{3}{4}\Big(\overline{a_d}^{pr}\big(C_{\phi2^2}a_d^{pr}(7g_1^2-9g_2^2)+C_{\phi1\phi2}\tilde\mu y_d^{pr}(6g_1^2-5g_2^2)\big)+$$
$$\tilde\mu\,\overline{y_d}^{pr}\big(\overline{C_{\phi1\phi2}}\,a_d^{pr}(6g_1^2-5g_2^2)+C_{\phi1^2}\tilde\mu y_d^{pr}(5g_1^2-3g_2^2)\big)\Big)\mathrm{LF}_{4,1,-1}[m_{\tilde q}^{p},m_{\tilde d}^{r}]+$$
$$\Big(\tilde\mu\,\overline{y_d}^{pr}\big(\overline{C_{\phi1\phi2}}\,a_d^{pr}(-3g_1^2+2g_2^2)+3C_{\phi1^2}\tilde\mu y_d^{pr}(-g_1^2+g_2^2)\big)+$$
$$\overline{a_d}^{pr}\big(3C_{\phi2^2}a_d^{pr}(-g_1^2+g_2^2)+C_{\phi1\phi2}\tilde\mu y_d^{pr}(-3g_1^2+2g_2^2)\big)\Big)\mathrm{LF}_{5,1,-2}[m_{\tilde q}^{p},m_{\tilde d}^{r}]-$$
$$3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,\mathrm{LF}_{1,1,0}[m_{\tilde q}^{p},m_{\tilde q}^{s}]+\frac{1}{4}(g_1^2+3g_2^2)(\overline{a_u}^{pr}a_u^{pr}-\tilde\mu^2\,\overline{y_u}^{pr}y_u^{pr})$$
$$\mathrm{LF}_{2,1,0}[m_{\tilde q}^{p},m_{\tilde u}^{r}]+\frac{1}{4}\Big(\overline{a_u}^{pr}\big(C_{\phi1^2}a_u^{pr}(g_1^2+3g_2^2)+2C_{\phi1\phi2}\tilde\mu g_2^2y_u^{pr}\big)+$$
$$\tilde\mu\,\overline{y_u}^{pr}\big(2\,\overline{C_{\phi1\phi2}}\,g_2^2a_u^{pr}-C_{\phi2^2}\tilde\mu y_u^{pr}(g_1^2+3g_2^2)\big)\Big)\mathrm{LF}_{2,2,0}[m_{\tilde q}^{p},m_{\tilde u}^{r}]-$$
$$\frac{1}{4}\tilde\mu g_2^2(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{pr}a_u^{pr}+C_{\phi1\phi2}\,y_u^{pr}\,\overline{a_u}^{pr})\mathrm{LF}_{3,2,-1}[m_{\tilde q}^{p},m_{\tilde u}^{r}]+$$
$$\frac{1}{4}(\overline{a_u}^{pr}a_u^{pr}(-7g_1^2+3g_2^2)+\tilde\mu^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+3g_2^2))\mathrm{LF}_{2,1,0}[m_{\tilde u}^{r},m_{\tilde q}^{p}]+$$
$$\frac{3}{4}(g_1^2-g_2^2)(\overline{a_u}^{pr}a_u^{pr}+\tilde\mu^2\,\overline{y_u}^{pr}y_u^{pr})\mathrm{LF}_{3,1,-1}[m_{\tilde u}^{r},m_{\tilde q}^{p}]+$$
$$\frac{1}{2}\Big(\overline{a_u}^{pr}\big(C_{\phi1^2}a_u^{pr}(-7g_1^2+3g_2^2)+C_{\phi1\phi2}\tilde\mu y_u^{pr}(-3g_1^2+2g_2^2)\big)+$$
$$\tilde\mu\,\overline{y_u}^{pr}\big(\overline{C_{\phi1\phi2}}\,a_u^{pr}(-3g_1^2+2g_2^2)+C_{\phi2^2}\tilde\mu y_u^{pr}(g_1^2+3g_2^2)\big)\Big)\mathrm{LF}_{3,1,0}[m_{\tilde u}^{r},m_{\tilde q}^{p}]+$$
$$\frac{1}{4}\Big(-\overline{a_u}^{pr}\big(C_{\phi1^2}a_u^{pr}(g_1^2+3g_2^2)+2C_{\phi1\phi2}\tilde\mu g_2^2y_u^{pr}\big)+$$
$$\tilde\mu\,\overline{y_u}^{pr}\big(-2\,\overline{C_{\phi1\phi2}}\,g_2^2a_u^{pr}+C_{\phi2^2}\tilde\mu y_u^{pr}(g_1^2+3g_2^2)\big)\Big)\mathrm{LF}_{3,2,-1}[m_{\tilde u}^{r},m_{\tilde q}^{p}]+$$
$$\frac{3}{2}\Big(\tilde\mu\,\overline{y_u}^{pr}\big(\overline{C_{\phi1\phi2}}\,a_u^{pr}(3g_1^2-2g_2^2)+C_{\phi2^2}\tilde\mu y_u^{pr}(g_1^2-3g_2^2)\big)+$$
$$\overline{a_u}^{pr}\big(C_{\phi1^2}a_u^{pr}(5g_1^2-3g_2^2)+C_{\phi1\phi2}\tilde\mu y_u^{pr}(3g_1^2-2g_2^2)\big)\Big)\mathrm{LF}_{4,1,-1}[m_{\tilde u}^{r},m_{\tilde q}^{p}]+$$
$$\Big(\tilde\mu\,\overline{y_u}^{pr}\big(\overline{C_{\phi1\phi2}}\,a_u^{pr}(-3g_1^2+2g_2^2)+3C_{\phi2^2}\tilde\mu y_u^{pr}(-g_1^2+g_2^2)\big)+$$
$$\overline{a_u}^{pr}\big(3C_{\phi1^2}a_u^{pr}(-g_1^2+g_2^2)+C_{\phi1\phi2}\tilde\mu y_u^{pr}(-3g_1^2+2g_2^2)\big)\Big)\mathrm{LF}_{5,1,-2}[m_{\tilde u}^{r},m_{\tilde q}^{p}]-$$
$$3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{2,1,0}[m_{\tilde u}^{t},m_{\tilde d}^{r}]+3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}$$
$$(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{3,1,-1}[m_{\tilde u}^{t},m_{\tilde d}^{r}]+\frac{3}{4}(g_1^4-g_1^2g_2^2)\mathrm{LF}_{2,1,-1}[\tilde\mu,m_1]+$$
$$\frac{1}{2}(-g_1^4+g_1^2g_2^2)\mathrm{LF}_{3,1,-2}[\tilde\mu,m_1]+g_1^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\mathrm{LF}_{3,1,-1}[\tilde\mu,m_1]-$$
$$\frac{1}{2}m_1\tilde\mu g_1^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(g_1^2-g_2^2)\mathrm{LF}_{3,1,0}[\tilde\mu,m_1]+\frac{3}{4}g_1^4(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{3,2,-2}[\tilde\mu,m_1]+$$
$$\frac{1}{4}m_1g_1^4\big(3m_1(C_{\phi1^2}+C_{\phi2^2})-4\tilde\mu(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})\big)\mathrm{LF}_{3,2,-1}[\tilde\mu,m_1]-$$
$$2g_1^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\mathrm{LF}_{4,1,-2}[\tilde\mu,m_1]+$$
$$\frac{1}{4}m_1\tilde\mu g_1^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(6g_1^2-5g_2^2)\mathrm{LF}_{4,1,-1}[\tilde\mu,m_1]-\frac{1}{2}g_1^4(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{4,2,-3}[\tilde\mu,m_1]+$$
$$\frac{1}{2}m_1g_1^4\big(-m_1(C_{\phi1^2}+C_{\phi2^2})+2\tilde\mu(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})\big)\mathrm{LF}_{4,2,-2}[\tilde\mu,m_1]+$$
$$g_1^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\mathrm{LF}_{5,1,-3}[\tilde\mu,m_1]-$$
$$\frac{1}{3}m_1\tilde\mu g_1^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(3g_1^2-2g_2^2)\mathrm{LF}_{5,1,-2}[\tilde\mu,m_1]+\frac{9}{4}g_2^2(g_1^2-g_2^2)\mathrm{LF}_{2,1,-1}[\tilde\mu,m_2]+$$
$$\frac{3}{2}g_2^2(-g_1^2+g_2^2)\mathrm{LF}_{3,1,-2}[\tilde\mu,m_2]+3g_2^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\mathrm{LF}_{3,1,-1}[\tilde\mu,m_2]+$$
$$\frac{1}{6}m_2\tilde\mu g_2^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(-9g_1^2+5g_2^2)\mathrm{LF}_{3,1,0}[\tilde\mu,m_2]+\frac{15}{4}g_2^4(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{3,2,-2}[\tilde\mu,m_2]+$$
$$\frac{1}{12}m_2g_2^4\big(9m_2(C_{\phi1^2}+C_{\phi2^2})-56\tilde\mu(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})\big)\mathrm{LF}_{3,2,-1}[\tilde\mu,m_2]+$$
$$6g_2^2(C_{\phi1^2}+C_{\phi2^2})(-g_1^2+g_2^2)\mathrm{LF}_{4,1,-2}[\tilde\mu,m_2]+$$
$$\frac{1}{4}m_2\tilde\mu g_2^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(18g_1^2-11g_2^2)\mathrm{LF}_{4,1,-1}[\tilde\mu,m_2]-\frac{5}{2}g_2^4(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{4,2,-3}[\tilde\mu,m_2]+$$
$$\frac{1}{2}g_2^4\big(-m_2^2(C_{\phi1^2}+C_{\phi2^2})+8m_2\tilde\mu(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})-2\tilde\mu^2(C_{\phi1^2}+C_{\phi2^2})\big)\mathrm{LF}_{4,2,-2}[\tilde\mu,m_2]+$$
$$3g_2^2(C_{\phi1^2}+C_{\phi2^2})(g_1^2-g_2^2)\mathrm{LF}_{5,1,-3}[\tilde\mu,m_2]+$$
$$m_2\tilde\mu g_2^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(-3g_1^2+2g_2^2)\mathrm{LF}_{5,1,-2}[\tilde\mu,m_2]+$$
$$\frac{3}{2}\tilde\mu g_1^2g_2^2\big(-(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(3m_1+m_2)+4\tilde\mu(C_{\phi1^2}+C_{\phi2^2})\big)\mathrm{LF}_{2,2,1,-1}[m_2,\tilde\mu,m_1]+$$
$$2\tilde\mu g_1^2g_2^2\big((\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(m_1+m_2)-2\tilde\mu(C_{\phi1^2}+C_{\phi2^2})\big)\mathrm{LF}_{3,2,1,-2}[m_2,\tilde\mu,m_1]-$$
$$3\tilde\mu^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde d}^{r},m_{\tilde d}^{t},m_{\tilde q}^{s}]+$$
$$\frac{3}{4}\tilde\mu\,\overline{y_d}^{st}y_d^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}-C_{\phi1\phi2}\,y_d^{pt}\,\overline{a_d}^{pr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde d}^{r},m_{\tilde q}^{p},m_{\tilde d}^{t}]-$$
$$3\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_d}^{pr}a_d^{sr}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde d}^{r},m_{\tilde q}^{p},m_{\tilde q}^{s}]+$$
$$3y_d^{sr}\,\overline{y_u}^{st}(-\overline{a_d}^{pr}a_u^{pt}+\tilde\mu^2\,\overline{y_d}^{pr}y_u^{pt})\mathrm{LF}_{1,1,1,0}[m_{\tilde d}^{r},m_{\tilde q}^{p},m_{\tilde u}^{t}]+$$
$$3\tilde\mu^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde d}^{r},m_{\tilde q}^{s},m_{\tilde u}^{t}]+$$
$$\frac{3}{4}\tilde\mu\,\overline{y_d}^{st}y_d^{sr}(-\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}+C_{\phi1\phi2}\,y_d^{pt}\,\overline{a_d}^{pr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde d}^{t},m_{\tilde q}^{p},m_{\tilde d}^{r}]-$$
$$3\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde d}^{t},m_{\tilde q}^{p},m_{\tilde u}^{r}]-\tilde\mu^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde e}^{r},m_{\tilde e}^{t},m_{\tilde l}^{s}]+$$
$$\frac{1}{4}\tilde\mu\,\overline{y_e}^{st}y_e^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}-C_{\phi1\phi2}\,y_e^{pt}\,\overline{a_e}^{pr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde e}^{r},m_{\tilde l}^{p},m_{\tilde e}^{t}]+$$
$$\frac{1}{4}\tilde\mu\,\overline{y_e}^{st}y_e^{sr}(-\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}+C_{\phi1\phi2}\,y_e^{pt}\,\overline{a_e}^{pr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde e}^{t},m_{\tilde l}^{p},m_{\tilde e}^{r}]-$$
$$\frac{1}{2}\tilde\mu\,\overline{y_e}^{st}y_e^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}+C_{\phi1\phi2}\,y_e^{pt}\,\overline{a_e}^{pr})\mathrm{LF}_{2,1,1,0}[m_{\tilde l}^{p},m_{\tilde e}^{r},m_{\tilde e}^{t}]+$$
$$\frac{1}{2}\tilde\mu\,\overline{y_e}^{st}y_e^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}+C_{\phi1\phi2}\,y_e^{pt}\,\overline{a_e}^{pr})\mathrm{LF}_{3,1,1,-1}[m_{\tilde l}^{p},m_{\tilde e}^{r},m_{\tilde e}^{t}]-$$
$$\frac{1}{4}C_{\phi1\phi2}\tilde\mu\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\overline{a_e}^{pr}\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde l}^{p},m_{\tilde l}^{s},m_{\tilde e}^{r}]-$$
$$\frac{1}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde l}^{p},m_{\tilde l}^{s},m_{\tilde e}^{t}]-$$
$$C_{\phi1^2}\tilde\mu^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde l}^{s},m_{\tilde e}^{r},m_{\tilde e}^{t}]+$$
$$C_{\phi1^2}\tilde\mu^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde l}^{s},m_{\tilde e}^{r},m_{\tilde e}^{t}]-$$
$$\frac{3}{2}\tilde\mu\,\overline{y_d}^{st}y_d^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}+C_{\phi1\phi2}\,y_d^{pt}\,\overline{a_d}^{pr})\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde d}^{t}]+$$
$$\frac{3}{2}\tilde\mu\,\overline{y_d}^{st}y_d^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}+C_{\phi1\phi2}\,y_d^{pt}\,\overline{a_d}^{pr})\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde d}^{t}]-$$
$$3\,\overline{y_u}^{st}y_u^{pt}a_d^{sr}(C_{\phi2^2}\overline{a_d}^{pr}+\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_d}^{pr})\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde q}^{s}]+$$
$$3\,\overline{y_u}^{st}y_u^{pt}a_d^{sr}(C_{\phi2^2}\overline{a_d}^{pr}+\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_d}^{pr})\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde q}^{s}]+3y_d^{sr}\,\overline{y_u}^{st}$$
$$\big(\tilde\mu\,\overline{y_d}^{pr}(-\overline{C_{\phi1\phi2}}\,a_u^{pt}+C_{\phi1^2}\tilde\mu y_u^{pt})+\overline{a_d}^{pr}(-C_{\phi2^2}a_u^{pt}+C_{\phi1\phi2}\tilde\mu y_u^{pt})\big)\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde u}^{t}]+$$
$$3y_d^{sr}\,\overline{y_u}^{st}\big(\tilde\mu\,\overline{y_d}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{pt}-C_{\phi1^2}\tilde\mu y_u^{pt})+\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde\mu y_u^{pt})\big)$$
$$\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde u}^{t}]-3\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde\mu y_d^{pt})\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p},m_{\tilde d}^{t},m_{\tilde u}^{r}]+$$
$$3\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde\mu y_d^{pt})\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p},m_{\tilde d}^{t},m_{\tilde u}^{r}]+$$
$$\frac{3}{4}\overline{a_d}^{pr}(4C_{\phi2^2}\overline{y_u}^{st}y_u^{pt}a_d^{sr}-C_{\phi1\phi2}\tilde\mu\,\overline{y_d}^{st}y_d^{pt}y_d^{sr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde d}^{r}]-$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde d}^{t}]-$$
$$3\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{r}]+$$
$$\frac{3}{4}C_{\phi1\phi2}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}(\overline{y_d}^{st}y_d^{pt}-\overline{y_u}^{st}y_u^{pt})\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{r}]+$$
$$\frac{3}{4}C_{\phi1\phi2}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}(-\overline{y_d}^{st}y_d^{pt}+\overline{y_u}^{st}y_u^{pt})\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{r}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(\overline{y_d}^{pr}y_d^{sr}-\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{t}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(-\overline{y_d}^{pr}y_d^{sr}+\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{t}]+$$
$$3\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde\mu y_d^{pt})\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p},m_{\tilde u}^{r},m_{\tilde d}^{t}]+$$
$$\frac{3}{8}C_{\phi1\phi2}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}(\overline{y_d}^{st}y_d^{pt}+\overline{y_u}^{st}y_u^{pt})\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p},m_{\tilde u}^{r},m_{\tilde q}^{s}]+3y_d^{sr}\,\overline{y_u}^{st}$$
$$\big(\tilde\mu\,\overline{y_d}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{pt}-C_{\phi1^2}\tilde\mu y_u^{pt})+\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde\mu y_u^{pt})\big)\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p},m_{\tilde u}^{t},m_{\tilde d}^{r}]+$$
$$\frac{3}{8}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(\overline{y_d}^{pr}y_d^{sr}+\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p},m_{\tilde u}^{t},m_{\tilde q}^{s}]-$$
$$3C_{\phi1^2}\tilde\mu^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde d}^{t}]+$$
$$3C_{\phi1^2}\tilde\mu^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde d}^{t}]-$$
$$3\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_d}^{pr}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde\mu y_d^{sr})\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde q}^{p}]+$$
$$3\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_d}^{pr}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde\mu y_d^{sr})\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde q}^{p}]+$$
$$3\tilde\mu\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde\mu y_d^{sr})\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde u}^{t}]-$$
$$3\tilde\mu\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde\mu y_d^{sr})\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde u}^{t}]+$$
$$\frac{9}{4}\tilde\mu(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}a_d^{sr}+C_{\phi1\phi2}\,y_d^{pr}\,\overline{y_u}^{pt}y_u^{st}\,\overline{a_d}^{sr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde d}^{r}]+$$
$$\frac{3}{4}C_{\phi1\phi2}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}(\overline{y_d}^{st}y_d^{pt}-\overline{y_u}^{st}y_u^{pt})\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde u}^{r}]+$$
$$\frac{3}{4}C_{\phi1\phi2}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}(-\overline{y_d}^{st}y_d^{pt}+\overline{y_u}^{st}y_u^{pt})\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde u}^{r}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(\overline{y_d}^{pr}y_d^{sr}-\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde u}^{t}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(-\overline{y_d}^{pr}y_d^{sr}+\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde u}^{t}]+$$
$$\frac{3}{8}C_{\phi1\phi2}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}(-3\,\overline{y_d}^{st}y_d^{pt}+\overline{y_u}^{st}y_u^{pt})\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{s},m_{\tilde u}^{r},m_{\tilde q}^{p}]-$$
$$3\tilde\mu^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde q}^{s},m_{\tilde u}^{r},m_{\tilde u}^{t}]-$$
$$3\tilde\mu\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde\mu y_d^{sr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{s},m_{\tilde u}^{t},m_{\tilde d}^{r}]+$$
$$\frac{3}{8}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(-3\,\overline{y_d}^{pr}y_d^{sr}+\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{s},m_{\tilde u}^{t},m_{\tilde q}^{p}]-$$
$$3\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{r},m_{\tilde d}^{t},m_{\tilde q}^{p}]+$$
$$3\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{r},m_{\tilde d}^{t},m_{\tilde q}^{p}]-$$
$$\frac{3}{2}\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}(2C_{\phi1^2}a_u^{sr}+C_{\phi1\phi2}\tilde\mu y_u^{sr})\mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{r},m_{\tilde q}^{p},m_{\tilde q}^{s}]+$$
$$\frac{3}{2}\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}(2C_{\phi1^2}a_u^{sr}+C_{\phi1\phi2}\tilde\mu y_u^{sr})\mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{r},m_{\tilde q}^{p},m_{\tilde q}^{s}]-$$
$$3C_{\phi1\phi2}\tilde\mu\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\overline{a_u}^{pr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{r},m_{\tilde q}^{p},m_{\tilde u}^{t}]+$$
$$3C_{\phi1\phi2}\tilde\mu\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\overline{a_u}^{pr}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{r},m_{\tilde q}^{p},m_{\tilde u}^{t}]-$$
$$3C_{\phi2^2}\tilde\mu^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{r},m_{\tilde q}^{s},m_{\tilde u}^{t}]+$$
$$3C_{\phi2^2}\tilde\mu^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{r},m_{\tilde q}^{s},m_{\tilde u}^{t}]+$$
$$\frac{3}{2}\tilde\mu\,\overline{y_u}^{st}y_u^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{pr}a_u^{pt}+C_{\phi1\phi2}\,y_u^{pt}\,\overline{a_u}^{pr})\mathrm{LF}_{2,2,1,-1}[m_{\tilde u}^{r},m_{\tilde u}^{t},m_{\tilde q}^{p}]+$$
$$3C_{\phi2^2}\tilde\mu^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde u}^{r},m_{\tilde u}^{t},m_{\tilde q}^{s}]-$$
$$3y_d^{sr}\,\overline{y_u}^{st}(C_{\phi1^2}+C_{\phi2^2})(\overline{a_d}^{pr}a_u^{pt}-\tilde\mu^2\,\overline{y_d}^{pr}y_u^{pt})\mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{t},m_{\tilde d}^{r},m_{\tilde q}^{p}]+$$
$$3y_d^{sr}\,\overline{y_u}^{st}(C_{\phi1^2}+C_{\phi2^2})(\overline{a_d}^{pr}a_u^{pt}-\tilde\mu^2\,\overline{y_d}^{pr}y_u^{pt})\mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{t},m_{\tilde d}^{r},m_{\tilde q}^{p}]+$$
$$3\tilde\mu^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{t},m_{\tilde d}^{r},m_{\tilde q}^{s}]-$$
$$3\tilde\mu^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{t},m_{\tilde d}^{r},m_{\tilde q}^{s}]-$$
$$\frac{3}{2}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}a_u^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{t},m_{\tilde q}^{p},m_{\tilde q}^{s}]+$$
$$\frac{3}{2}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}a_u^{pt}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{t},m_{\tilde q}^{p},m_{\tilde q}^{s}]-$$
$$3\,\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{t},m_{\tilde q}^{p},m_{\tilde u}^{r}]+$$
$$3\,\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{t},m_{\tilde q}^{p},m_{\tilde u}^{r}]-$$
$$3C_{\phi2^2}\tilde\mu^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{t},m_{\tilde q}^{s},m_{\tilde u}^{r}]+$$
$$3C_{\phi2^2}\tilde\mu^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{t},m_{\tilde q}^{s},m_{\tilde u}^{r}]-$$
$$g_1^2g_2^2\,\mathrm{LF}_{2,1,1,-2}[\tilde\mu,m_1,m_2]+g_1^2g_2^2(-m_1m_2+4\tilde\mu^2)\mathrm{LF}_{2,1,1,-1}[\tilde\mu,m_1,m_2]+$$
$$3m_1\tilde\mu g_1^2g_2^2(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})\mathrm{LF}_{2,1,1,0}[\tilde\mu,m_1,m_2]-\frac{3}{2}g_1^2g_2^2(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{3,1,1,-2}[\tilde\mu,m_1,m_2]+$$
$$\frac{1}{4}g_1^2g_2^2\big(-6m_1m_2(C_{\phi1^2}+C_{\phi2^2})-\tilde\mu(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(31m_1-5m_2)+24\tilde\mu^2(C_{\phi1^2}+C_{\phi2^2})\big)$$
$$\mathrm{LF}_{3,1,1,-1}[\tilde\mu,m_1,m_2]+g_1^2g_2^2(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{4,1,1,-3}[\tilde\mu,m_1,m_2]+$$
$$\frac{1}{2}g_1^2g_2^2\big(2m_1m_2(C_{\phi1^2}+C_{\phi2^2})+3\tilde\mu(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})(3m_1-m_2)-10\tilde\mu^2(C_{\phi1^2}+C_{\phi2^2})\big)$$
$$\mathrm{LF}_{4,1,1,-2}[\tilde\mu,m_1,m_2]+4\tilde\mu g_1^2g_2^2\big(m_1(\overline{C_{\phi1\phi2}}+C_{\phi1\phi2})-\tilde\mu(C_{\phi1^2}+C_{\phi2^2})\big)\mathrm{LF}_{3,2,1,-2}[\tilde\mu,m_2,m_1]-$$
$$3\tilde\mu^2\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r},m_{\tilde d}^{t},m_{\tilde q}^{p},m_{\tilde q}^{s}]+$$
$$\frac{3}{4}\tilde\mu^3\,\overline{y_d}^{st}y_d^{sr}(-\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}+C_{\phi1\phi2}\,y_d^{pt}\,\overline{a_d}^{pr})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{r},m_{\tilde q}^{p},m_{\tilde d}^{t},m_{\tilde q}^{s}]-$$
$$3\big(\tilde\mu^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}a_u^{pt}(a_d^{sr}\,\overline{a_u}^{st}-\tilde\mu^2y_d^{sr}\,\overline{y_u}^{st})\big)\mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r},m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{t}]+$$
$$\frac{3}{4}\tilde\mu\,\overline{a_d}^{pr}a_d^{pt}(-\overline{C_{\phi1\phi2}}\,\overline{y_d}^{st}a_d^{sr}+C_{\phi1\phi2}\,y_d^{sr}\,\overline{a_d}^{st})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{r},m_{\tilde q}^{s},m_{\tilde d}^{t},m_{\tilde q}^{p}]+$$
$$\frac{3}{4}\tilde\mu^3\,\overline{y_d}^{st}y_d^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}a_d^{pt}-C_{\phi1\phi2}\,y_d^{pt}\,\overline{a_d}^{pr})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{t},m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde q}^{s}]+$$
$$3\tilde\mu^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{t},m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{r}]+$$
$$\frac{3}{4}\tilde\mu\,\overline{a_d}^{pr}a_d^{pt}(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{st}a_d^{sr}-C_{\phi1\phi2}\,y_d^{sr}\,\overline{a_d}^{st})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{t},m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde q}^{p}]-$$
$$\tilde\mu^2\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde e}^{r},m_{\tilde e}^{t},m_{\tilde l}^{p},m_{\tilde l}^{s}]+$$
$$\frac{1}{4}\tilde\mu^3\,\overline{y_e}^{st}y_e^{sr}(-\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}+C_{\phi1\phi2}\,y_e^{pt}\,\overline{a_e}^{pr})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{r},m_{\tilde l}^{p},m_{\tilde e}^{t},m_{\tilde l}^{s}]+$$
$$\frac{1}{4}\tilde\mu\,\overline{a_e}^{pr}a_e^{pt}(-\overline{C_{\phi1\phi2}}\,\overline{y_e}^{st}a_e^{sr}+C_{\phi1\phi2}\,y_e^{sr}\,\overline{a_e}^{st})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{r},m_{\tilde l}^{s},m_{\tilde e}^{t},m_{\tilde l}^{p}]+$$
$$\frac{1}{4}\tilde\mu^3\,\overline{y_e}^{st}y_e^{sr}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr}a_e^{pt}-C_{\phi1\phi2}\,y_e^{pt}\,\overline{a_e}^{pr})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{t},m_{\tilde l}^{p},m_{\tilde e}^{r},m_{\tilde l}^{s}]+$$
$$\frac{1}{4}\tilde\mu\,\overline{a_e}^{pr}a_e^{pt}(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{st}a_e^{sr}-C_{\phi1\phi2}\,y_e^{sr}\,\overline{a_e}^{st})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{t},m_{\tilde l}^{s},m_{\tilde e}^{r},m_{\tilde l}^{p}]-\frac{1}{2}\tilde\mu^2\,\overline{y_e}^{st}y_e^{sr}$$
$$\big(\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_e}^{pr}a_e^{pt}+\overline{a_e}^{pr}(2C_{\phi2^2}a_e^{pt}+C_{\phi1\phi2}\tilde\mu y_e^{pt})\big)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde l}^{p},m_{\tilde e}^{r},m_{\tilde e}^{t},m_{\tilde l}^{s}]+\frac{1}{2}\tilde\mu^2\,\overline{y_e}^{st}$$
$$y_e^{sr}\big(\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_e}^{pr}a_e^{pt}+\overline{a_e}^{pr}(2C_{\phi2^2}a_e^{pt}+C_{\phi1\phi2}\tilde\mu y_e^{pt})\big)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde l}^{p},m_{\tilde e}^{r},m_{\tilde e}^{t},m_{\tilde l}^{s}]-$$
$$\frac{1}{2}\tilde\mu\,\overline{a_e}^{pr}a_e^{pt}\big(C_{\phi1\phi2}\,y_e^{sr}\,\overline{a_e}^{st}+\overline{y_e}^{st}(\overline{C_{\phi1\phi2}}\,a_e^{sr}+2C_{\phi1^2}\tilde\mu y_e^{sr})\big)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde l}^{s},m_{\tilde e}^{r},m_{\tilde e}^{t},m_{\tilde l}^{p}]+$$
$$\frac{1}{2}\tilde\mu\,\overline{a_e}^{pr}a_e^{pt}\big(C_{\phi1\phi2}\,y_e^{sr}\,\overline{a_e}^{st}+\overline{y_e}^{st}(\overline{C_{\phi1\phi2}}\,a_e^{sr}+2C_{\phi1^2}\tilde\mu y_e^{sr})\big)$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde l}^{s},m_{\tilde e}^{r},m_{\tilde e}^{t},m_{\tilde l}^{p}]-\frac{3}{2}\tilde\mu^2\,\overline{y_d}^{st}y_d^{sr}$$
$$\big(\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_d}^{pr}a_d^{pt}+\overline{a_d}^{pr}(2C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde\mu y_d^{pt})\big)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde d}^{t},m_{\tilde q}^{s}]+$$
$$\frac{3}{2}\tilde\mu^2\,\overline{y_d}^{st}y_d^{sr}\big(\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_d}^{pr}a_d^{pt}+\overline{a_d}^{pr}(2C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde\mu y_d^{pt})\big)$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde d}^{t},m_{\tilde q}^{s}]+3\big(\tilde\mu^3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_u^{pt}-C_{\phi1^2}\tilde\mu y_u^{pt})+$$
$$\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde\mu y_u^{pt})(-a_d^{sr}\,\overline{a_u}^{st}+\tilde\mu^2y_d^{sr}\,\overline{y_u}^{st})\big)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde q}^{s},m_{\tilde u}^{t}]+$$
$$3\big(\tilde\mu^3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(-\overline{C_{\phi1\phi2}}\,a_u^{pt}+C_{\phi1^2}\tilde\mu y_u^{pt})+\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde\mu y_u^{pt})$$
$$(a_d^{sr}\,\overline{a_u}^{st}-\tilde\mu^2y_d^{sr}\,\overline{y_u}^{st})\big)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde q}^{s},m_{\tilde u}^{t}]+$$
$$3\tilde\mu^2\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde\mu y_d^{pt})\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p},m_{\tilde d}^{t},m_{\tilde q}^{s},m_{\tilde u}^{r}]-$$
$$3\tilde\mu^2\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde\mu y_d^{pt})\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p},m_{\tilde d}^{t},m_{\tilde q}^{s},m_{\tilde u}^{r}]-$$
$$\frac{3}{4}\big(C_{\phi1\phi2}\tilde\mu y_u^{st}\,\overline{a_d}^{sr}a_d^{pr}\,\overline{a_u}^{pt}+\tilde\mu^3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_u^{pt}-4C_{\phi1^2}\tilde\mu y_u^{pt})+\overline{a_d}^{pr}a_u^{pt}$$
$$\big(\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_d^{sr}-\overline{a_u}^{st}(4C_{\phi2^2}a_d^{sr}+3C_{\phi1\phi2}\tilde\mu y_d^{sr})\big)\big)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde u}^{t}]-$$
$$\frac{3}{4}\tilde\mu\,\overline{y_d}^{st}\,\overline{a_u}^{pr}(-3\,\overline{C_{\phi1\phi2}}\,a_d^{pt}a_u^{sr}+C_{\phi1\phi2}\tilde\mu^2y_d^{pt}y_u^{sr})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde d}^{t},m_{\tilde u}^{r}]-$$
$$3\tilde\mu^2\,\overline{y_u}^{st}y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{r},m_{\tilde u}^{t}]-$$
$$\frac{3}{4}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}\big(C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde\mu y_u^{pt})\big)$$
$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{r},m_{\tilde u}^{t}]+\frac{3}{4}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}$$
$$\big(C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde\mu y_u^{pt})\big)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{r},m_{\tilde u}^{t}]-$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(\overline{a_u}^{pr}a_u^{sr}+\tilde\mu^2\,\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{t},m_{\tilde u}^{r}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(\overline{a_u}^{pr}a_u^{sr}+\tilde\mu^2\,\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{t},m_{\tilde u}^{r}]-$$
$$3\tilde\mu^2\,\overline{y_d}^{st}y_u^{sr}\,\overline{a_u}^{pr}(C_{\phi2^2}a_d^{pt}+C_{\phi1\phi2}\tilde\mu y_d^{pt})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p},m_{\tilde u}^{r},m_{\tilde d}^{t},m_{\tilde q}^{s}]+$$
$$\frac{3}{8}\big(-\overline{C_{\phi1\phi2}}\,\tilde\mu^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+$$
$$\tilde\mu\,\overline{a_u}^{pr}\big(3\,\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(4C_{\phi2^2}a_u^{pt}+C_{\phi1\phi2}\tilde\mu y_u^{pt}))\big)\big)$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p},m_{\tilde u}^{r},m_{\tilde q}^{s},m_{\tilde u}^{t}]+3\big(\tilde\mu^3\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}(-\overline{C_{\phi1\phi2}}\,a_u^{pt}+C_{\phi1^2}\tilde\mu y_u^{pt})+$$
$$\overline{a_d}^{pr}(C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde\mu y_u^{pt})(a_d^{sr}\,\overline{a_u}^{st}-\tilde\mu^2y_d^{sr}\,\overline{y_u}^{st})\big)$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p},m_{\tilde u}^{t},m_{\tilde d}^{r},m_{\tilde q}^{s}]+\frac{3}{8}\big(\overline{C_{\phi1\phi2}}\,\tilde\mu^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+$$
$$\tilde\mu\,\overline{a_u}^{pr}\big(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(3C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(4C_{\phi2^2}a_u^{pt}-C_{\phi1\phi2}\tilde\mu y_u^{pt}))\big)\big)$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p},m_{\tilde u}^{t},m_{\tilde q}^{s},m_{\tilde u}^{r}]-\frac{3}{2}\tilde\mu\,\overline{a_d}^{pr}a_d^{pt}$$
$$\big(C_{\phi1\phi2}\,y_d^{sr}\,\overline{a_d}^{st}+\overline{y_d}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+2C_{\phi1^2}\tilde\mu y_d^{sr})\big)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde d}^{t},m_{\tilde q}^{p}]+$$
$$\frac{3}{2}\tilde\mu\,\overline{a_d}^{pr}a_d^{pt}\big(C_{\phi1\phi2}\,y_d^{sr}\,\overline{a_d}^{st}+\overline{y_d}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+2C_{\phi1^2}\tilde\mu y_d^{sr})\big)$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde d}^{t},m_{\tilde q}^{p}]-3\big(\tilde\mu^3\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde\mu y_d^{sr})+$$
$$\overline{a_d}^{pr}a_u^{pt}\big(-\tilde\mu\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde\mu y_d^{sr})+\overline{a_u}^{st}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde\mu y_d^{sr})\big)\big)$$
$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde q}^{p},m_{\tilde u}^{t}]+3\big(\tilde\mu^3\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde\mu y_d^{sr})+$$
$$\overline{a_d}^{pr}a_u^{pt}\big(-\tilde\mu\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde\mu y_d^{sr})+\overline{a_u}^{st}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde\mu y_d^{sr})\big)\big)$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s},m_{\tilde d}^{r},m_{\tilde q}^{p},m_{\tilde u}^{t}]+3\tilde\mu\,\overline{y_d}^{st}a_d^{pt}\,\overline{a_u}^{pr}(-\overline{C_{\phi1\phi2}}\,a_u^{sr}+C_{\phi1^2}\tilde\mu y_u^{sr})$$
$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s},m_{\tilde d}^{t},m_{\tilde q}^{p},m_{\tilde u}^{r}]+$$
$$3\tilde\mu\,\overline{y_d}^{st}a_d^{pt}\,\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{sr}-C_{\phi1^2}\tilde\mu y_u^{sr})\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s},m_{\tilde d}^{t},m_{\tilde q}^{p},m_{\tilde u}^{r}]+$$
$$\frac{9}{4}\tilde\mu^3(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}a_d^{sr}+C_{\phi1\phi2}\,y_d^{pr}\,\overline{y_u}^{pt}y_u^{st}\,\overline{a_d}^{sr})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde d}^{r},m_{\tilde u}^{t}]-$$
$$\frac{3}{4}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}\big(C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde\mu y_u^{pt})\big)$$
$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde u}^{r},m_{\tilde u}^{t}]+\frac{3}{4}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}$$
$$\big(C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde\mu y_u^{pt})\big)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde u}^{r},m_{\tilde u}^{t}]-$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(\overline{a_u}^{pr}a_u^{sr}+\tilde\mu^2\,\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde u}^{t},m_{\tilde u}^{r}]+$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde\mu\,\overline{y_u}^{st}a_u^{pt}(\overline{a_u}^{pr}a_u^{sr}+\tilde\mu^2\,\overline{y_u}^{pr}y_u^{sr})\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s},m_{\tilde q}^{p},m_{\tilde u}^{t},m_{\tilde u}^{r}]+$$
$$3\tilde\mu\,\overline{y_d}^{st}a_d^{pt}\,\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{sr}-C_{\phi1^2}\tilde\mu y_u^{sr})\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s},m_{\tilde u}^{r},m_{\tilde d}^{t},m_{\tilde q}^{p}]+$$
$$\frac{3}{8}\tilde\mu\big(3\,\overline{C_{\phi1\phi2}}\,\tilde\mu^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+$$
$$\overline{a_u}^{pr}\big(-\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(4C_{\phi1^2}a_u^{pt}+C_{\phi1\phi2}\tilde\mu y_u^{pt}))\big)\big)$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s},m_{\tilde u}^{r},m_{\tilde q}^{p},m_{\tilde u}^{t}]+3\big(\tilde\mu^3\,\overline{y_d}^{pr}\,\overline{y_u}^{st}y_u^{pt}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde\mu y_d^{sr})+$$
$$\overline{a_d}^{pr}a_u^{pt}\big(-\tilde\mu\,\overline{y_u}^{st}(\overline{C_{\phi1\phi2}}\,a_d^{sr}+C_{\phi1^2}\tilde\mu y_d^{sr})+\overline{a_u}^{st}(C_{\phi2^2}a_d^{sr}+C_{\phi1\phi2}\tilde\mu y_d^{sr})\big)\big)$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s},m_{\tilde u}^{t},m_{\tilde d}^{r},m_{\tilde q}^{p}]+\frac{3}{8}\big(\overline{C_{\phi1\phi2}}\,\tilde\mu^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+$$
$$\tilde\mu\,\overline{a_u}^{pr}\big(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(-C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(4C_{\phi1^2}a_u^{pt}+3C_{\phi1\phi2}\tilde\mu y_u^{pt}))\big)\big)$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s},m_{\tilde u}^{t},m_{\tilde q}^{p},m_{\tilde u}^{r}]+3\tilde\mu^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(C_{\phi1^2}+C_{\phi2^2})$$
$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{r},m_{\tilde d}^{t},m_{\tilde q}^{p},m_{\tilde q}^{s}]-$$
$$3\tilde\mu^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(C_{\phi1^2}+C_{\phi2^2})\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{r},m_{\tilde d}^{t},m_{\tilde q}^{p},m_{\tilde q}^{s}]-\frac{3}{2}\tilde\mu\,\overline{y_u}^{st}a_u^{pt}$$
$$\big(\overline{C_{\phi1\phi2}}\,\tilde\mu^2\,\overline{y_u}^{pr}y_u^{sr}+\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{sr}+\tilde\mu y_u^{sr}(C_{\phi1^2}+C_{\phi2^2}))\big)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{r},m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{t}]+$$
$$\frac{3}{2}\tilde\mu\,\overline{y_u}^{st}a_u^{pt}\big(\overline{C_{\phi1\phi2}}\,\tilde\mu^2\,\overline{y_u}^{pr}y_u^{sr}+\overline{a_u}^{pr}(\overline{C_{\phi1\phi2}}\,a_u^{sr}+\tilde\mu y_u^{sr}(C_{\phi1^2}+C_{\phi2^2}))\big)$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{r},m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{t}]+\frac{3}{4}\big(\overline{C_{\phi1\phi2}}\,\tilde\mu^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}+\tilde\mu\,\overline{a_u}^{pr}$$
$$\big(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}a_u^{pt}a_u^{sr}+y_u^{sr}(C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(2a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde\mu y_u^{pt}))\big)\big)$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde u}^{r},m_{\tilde u}^{t},m_{\tilde q}^{p},m_{\tilde q}^{s}]-3(C_{\phi1^2}+C_{\phi2^2})$$
$$\big(\tilde\mu^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}a_u^{pt}(a_d^{sr}\,\overline{a_u}^{st}-\tilde\mu^2y_d^{sr}\,\overline{y_u}^{st})\big)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{t},m_{\tilde d}^{r},m_{\tilde q}^{p},m_{\tilde q}^{s}]+$$
$$3(C_{\phi1^2}+C_{\phi2^2})\big(\tilde\mu^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}a_u^{pt}(a_d^{sr}\,\overline{a_u}^{st}-\tilde\mu^2y_d^{sr}\,\overline{y_u}^{st})\big)$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{t},m_{\tilde d}^{r},m_{\tilde q}^{p},m_{\tilde q}^{s}]-\frac{3}{2}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}$$
$$\big(C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde\mu y_u^{pt})\big)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{t},m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{r}]+$$
$$\frac{3}{2}\tilde\mu y_u^{sr}\,\overline{a_u}^{pr}\big(C_{\phi1\phi2}\,\overline{a_u}^{st}a_u^{pt}+\tilde\mu\,\overline{y_u}^{st}(a_u^{pt}(C_{\phi1^2}+C_{\phi2^2})+C_{\phi1\phi2}\tilde\mu y_u^{pt})\big)$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{t},m_{\tilde q}^{p},m_{\tilde q}^{s},m_{\tilde u}^{r}]\Big)$$